

FAQ - Frequently Asked Questions

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General Questions

What is CalDAV?

CalDAV (Calendaring Extensions to WebDAV) is an Internet standard allowing clients to access calendar information on a remote server. It's defined in RFC 4791 and widely supported by calendar servers like Nextcloud, ownCloud, Google Calendar, and Apple iCloud.

Why use CalDAV on ESP32?

CalDAV enables ESP32 devices to:

- Retrieve upcoming events for display on smart displays
- Trigger actions based on calendar events (home automation)
- Synchronize schedules with cloud services
- Create calendar-aware IoT devices

Which CalDAV servers are supported?

The library works with any RFC 4791 compliant CalDAV server, including:

- Nextcloud
- ownCloud
- Radicale
- Baikal
- SOGo
- Apple Calendar Server
- Google Calendar (with CalDAV API)

Configuration & Setup

How do I find my CalDAV server URL?

The server URL format varies by provider:

Nextcloud/ownCloud:

```
https://your-server.com/remote.php/dav/calendars/username
```

Radicale:

```
https://your-server.com/username
```

Baikal:

```
https://your-server.com/cal.php/calendars/username
```

For other servers, consult their documentation or check the calendar app settings.

What credentials do I need?

You need:

- Username (usually your login name or email)
- Password or app-specific password

NOTE

If your server uses two-factor authentication (2FA), you'll need to generate an app-specific password.

How do I generate an app-specific password?

Nextcloud:

1. Go to Settings → Security
2. Scroll to "Devices & sessions"
3. Enter an app name and click "Create new app password"
4. Copy the generated password

Google Calendar:

1. Go to Google Account settings
2. Security → 2-Step Verification → App passwords
3. Generate a new app password for "Other (custom name)"

Can I use a self-signed certificate?

The library uses ESP-IDF's certificate bundle by default. For self-signed certificates, you need to:

1. Disable certificate verification (not recommended for production)
2. Or add your certificate to the bundle
3. Or implement custom certificate validation

Example (disable verification - use only for testing):

```
// In caldav_client.cpp, modify HTTP config:  
_CalDAV_HTTP_Config.skip_cert_common_name_check = true;  
_CalDAV_HTTP_Config.crt_bundle_attach = NULL;
```

Connection Issues

I get CALDAV_ERROR_CONNECTION. What should I do?

Check these common issues:

1. **Network connectivity:** Ensure Wi-Fi is connected
2. **DNS resolution:** Verify the server hostname resolves correctly
3. **Firewall:** Check if firewall blocks outbound HTTPS (port 443)
4. **Server availability:** Test the URL in a web browser

Debug steps:

```
// Enable verbose logging
esp_log_level_set("CalDAV-Client", ESP_LOG_DEBUG);
esp_log_level_set("HTTP_CLIENT", ESP_LOG_DEBUG);
```

I get CALDAV_ERROR_HTTP with status 401

This indicates authentication failure. Check:

- Username and password are correct
- Account is not locked
- Two-factor authentication app password is used (if enabled)
- Server URL points to the correct endpoint

I get CALDAV_ERROR_TIMEOUT

The request is taking too long. Try:

- Increasing timeout: `config.TimeoutMs = 30000`; (30 seconds)
- Checking network latency
- Reducing the query time range for events
- Verifying server performance

The connection works in a browser but not on ESP32

This usually indicates:

- Certificate validation issues
- Missing or incorrect certificate bundle
- Server using older TLS version not supported by ESP32

Enable certificate bundle in sdkconfig:

```
CONFIG_MBEDTLS_CERTIFICATE_BUNDLE=y
CONFIG_MBEDTLS_CERTIFICATE_BUNDLE_DEFAULT_FULL=y
```

Calendar & Event Queries

No calendars are returned

If `CalDAV_Calendars_List()` returns 0 calendars:

1. Verify the server URL is correct
2. Check user has at least one calendar
3. Ensure user has permission to access calendars
4. Enable debug logging to inspect server response

No events are returned

If `CalDAV_Calendar_Events_List()` returns 0 events:

1. Verify events exist in the specified time range
2. Check calendar path is correct (use path from `CalDAV_Calendars_List()`)
3. Ensure time format is correct (YYYYMMDDTHH:mm:ssZ)
4. Try expanding the time range

How do I query all future events?

Use a far future end date:

```
CalDAV_Calendar_Events_List(  
    client,  
    &events,  
    &event_count,  
    calendar_path,  
    "20250101T000000Z", // Current or past date  
    "20501231T235959Z"  // Far future date  
);
```

Can I search for specific events?

The library doesn't support text search directly. Retrieve all events and filter locally:

```
for (size_t i = 0; i < event_count; i++) {  
    if (events[i].Summary && strstr(events[i].Summary, "Meeting")) {  
        // Found matching event  
    }  
}
```

What date/time format is required?

Use iCalendar format:

- **UTC time:** YYYYMMDDTHH:mm:ssZ (Z indicates UTC)
- **Local time:** YYYYMMDDTHH:mm:ss (no Z suffix)

Examples:

```
"20250315T143000Z" = March 15, 2025, 14:30:00 UTC  
"20250101T000000Z" = January 1, 2025, 00:00:00 UTC
```

How do I handle recurring events?

CalDAV servers typically expand recurring events within the queried time range. Each occurrence appears as a separate event with its own start/end time.

Events show weird times. Why?

The library returns times as received from the server. You may need to:

1. Parse the iCalendar format
2. Convert to local timezone
3. Handle **VALUE=DATE** for all-day events

Example parsing:

```
// For VALUE=DATE format (e.g., "20250210")  
if (strlen(events[i].StartTime) == 8) {  
    // All-day event  
    int year, month, day;  
    sscanf(events[i].StartTime, "%4d%2d%2d", &year, &month, &day);  
}
```

Memory Issues

I get CALDAV_ERROR_NO_MEM errors

The system is out of memory. Solutions:

1. **Enable PSRAM:**

```
CONFIG_ESP32_CALDAV_USEPSRAM=y
```

2. **Increase heap size:** Adjust partition table
3. **Reduce query range:** Fetch events in smaller time windows
4. **Free resources promptly:**

```
CalDAV_Events_Free(events, event_count);
```

```
CalDAV_Calendars_Free(&calendars);
```

5. **Check for memory leaks:** Ensure all allocated resources are freed

How much memory does the library use?

Typical usage:

- Client handle: ~100 bytes
- Per calendar: ~200 bytes + string data
- Per event: ~300 bytes + string data
- HTTP buffers: 4 KB (expandable)

For a calendar with 50 events averaging 100 bytes each, expect ~20 KB total.

Can I limit memory usage?

Yes, several strategies:

1. Query shorter time ranges
2. Process events in batches and free between queries
3. Limit the number of calendars queried
4. Use PSRAM for large allocations

Performance & Optimization

How long does a typical query take?

Approximate times (depends on network and server):

- Connection test: 1-2 seconds
- List calendars: 2-5 seconds
- Fetch events (10-50 events): 3-10 seconds

Can I make queries faster?

1. **Reduce timeout** (but risk timeouts on slow networks)
2. **Cache results** and refresh periodically
3. **Use narrower time ranges**
4. **Minimize calendar/event parsing**

How often should I sync?

Depends on your use case:

- Smart display: Every 5-15 minutes
- Home automation: Every hour or event-triggered
- Status indicators: Every 30 minutes

Balance freshness needs with server load and power consumption.

Should I use FreeRTOS tasks?

Yes, for periodic synchronization:

```
void caldav_sync_task(void *pvParameters)
{
    CalDAV_Client_t *client = (CalDAV_Client_t *)pvParameters;

    while (1) {
        // Sync calendars and events
        // ...

        vTaskDelay(pdMS_TO_TICKS(300000)); // 5 minutes
    }
}

xTaskCreate(caldav_sync_task, "caldav", 8192, client, 5, NULL);
```

Security

Is my password transmitted securely?

Yes, the library uses HTTPS (TLS/SSL) for all communications. Passwords are sent using HTTP Basic Authentication over the encrypted connection.

Should I hardcode credentials?

No! Use secure storage:

1. **NVS (Non-Volatile Storage):** For encrypted credential storage
2. **Secure Boot:** Protect firmware from tampering
3. **Flash Encryption:** Encrypt the entire flash

Can I use OAuth instead of passwords?

The current library version supports only HTTP Basic Authentication. OAuth support may be added in future versions.

For OAuth-enabled servers, generate an app-specific password as a workaround.

How do I store credentials securely?

Example using NVS:

```
#include "nvs_flash.h"
#include "nvs.h"

esp_err_t save_credentials(const char *user, const char *pass)
{
    nvs_handle_t handle;
    esp_err_t err = nvs_open("caldav", NVS_READWRITE, &handle);
    if (err != ESP_OK) return err;

    nvs_set_str(handle, "user", user);
    nvs_set_str(handle, "pass", pass);
    nvs_commit(handle);
    nvs_close(handle);

    return ESP_OK;
}
```

Troubleshooting

How do I enable debug logging?

```
#include <esp_log.h>

// In app_main() or before CalDAV calls:
esp_log_level_set("CalDAV-Client", ESP_LOG_DEBUG);
esp_log_level_set("HTTP_CLIENT", ESP_LOG_VERBOSE);
```

The library crashes. What should I check?

1. **Stack overflow:** Increase task stack size
2. **Heap corruption:** Enable heap tracing
3. **Null pointers:** Check all returned pointers

4. **Memory leaks:** Ensure `_Free()` functions are called

Enable heap debugging:

```
CONFIG_HEAP_POISONING_COMPREHENSIVE=y  
CONFIG_HEAP_TRACING=y
```

Server returns HTML instead of XML

This usually means:

- Wrong server URL (hitting a web page instead of CalDAV endpoint)
- Server redirecting to login page
- Authentication required but not provided

The library detects HTML and returns `CALDAV_ERROR_HTTP`.

Events are duplicated

Check if:

- You're querying overlapping time ranges
- The same event occurs in multiple calendars
- Server expands recurring events into the same period multiple times

Strange characters in event titles

This indicates encoding issues:

1. Ensure server uses UTF-8 encoding
2. Check terminal/display supports UTF-8
3. Server may be sending escaped characters

Development & Integration

Can I use this with Arduino framework?

Yes! The library works with PlatformIO and Arduino-ESP32 framework:

```
platform = espressif32  
framework = arduino  
lib_deps =  
    https://github.com/Kampi/ESP32-CalDAV.git
```

How do I integrate with LVGL?

The library provides data; you display it with LVGL:

```
void display_events_on_lvgl(CalDAV_Calendar_Event_t *events, size_t count)
{
    lv_obj_t *list = lv_list_create(lv_scr_act());

    for (size_t i = 0; i < count; i++) {
        lv_obj_t *item = lv_list_add_btn(list, LV_SYMBOL_CALENDAR,
                                         events[i].Summary);
    }
}
```

Can I modify the library?

Yes! The library is licensed under GPLv3:

- You can modify and redistribute
- You must keep the same license
- You must provide source code for modifications

How do I contribute?

1. Fork the repository
2. Create a feature branch
3. Make your changes
4. Submit a pull request

Or report issues and suggestions via GitHub Issues.

Is there example code?

Yes, check the [examples/](#) directory in the repository for:

- Basic connection test
- Calendar listing
- Event synchronization
- Full application examples

Platform-Specific

Does it work on ESP32-S2/S3/C3?

Yes, the library is compatible with:

- ESP32
- ESP32-S2
- ESP32-S3
- ESP32-C3

All variants with sufficient memory can run the library.

What about memory-constrained devices?

For devices with limited RAM:

1. Enable PSRAM if available
2. Query smaller time ranges
3. Process one calendar at a time
4. Free resources immediately after use

Minimum recommended: 100 KB free heap

Can I use it without Wi-Fi?

CalDAV requires network connectivity. Alternatives:

- Ethernet adapter
- LTE/cellular modem
- LoRaWAN (with custom gateway)

The library works with any network interface supported by ESP-IDF.

Support & Resources

Where can I get help?

- Email: DanielKampert@kampus-elektroecke.de
- GitHub Issues: <https://github.com/Kampi/ESP32-CalDAV/issues>
- ESP-IDF Forum: <https://esp32.com>

Where is the source code?

GitHub: <https://github.com/Kampi/ESP32-CalDAV>

How do I report bugs?

Open an issue on GitHub with:

1. ESP-IDF version
2. Hardware (ESP32 variant)
3. CalDAV server type and version
4. Complete error messages
5. Minimal code to reproduce

Is commercial support available?

Contact DanielKampert@kampus-elektroecke.de for commercial support inquiries.